

DEM 601 Introduction to Demographic Analysis

Formula sheet

COVERS

lectures 1 to 2

TERM

Fall 2026

Every equation the course has shown you, in the order it was shown.

Lecture 1 • Introduction to the Course and Basic Demographic Concepts

$$P_{n+k} = P_n + \underbrace{{}_k B_n - {}_k D_n}_{\text{natural increase}} + \underbrace{{}_k I_n - {}_k E_n}_{\text{net migration}}$$

population change

What makes a population change

Lecture 2 • Demographic Indicators and Person-Years-Lived

$$\text{indicator} = \frac{\text{numerator}}{\text{denominator}} \times k$$

What an indicator is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mean

$$M = L_0 + \left(\frac{P/2 - \sum f_0}{f_0} \right) i$$

The median of grouped data

$$\text{ratio} = \frac{a}{b} \times k$$

Ratio

$$\text{proportion} = \frac{a}{a+b}$$

Proportion

$$\text{percentage} = \frac{a}{a+b} \times 100$$

Percentage

$$\text{rate} = \frac{e}{PY} \times k$$

A rate needs time

$$CBR = \frac{\text{births in the year}}{\text{mid-year population}} \times 1000$$

Is it a true rate?

$$IMR = \frac{\text{deaths under age 1}}{\text{live births in the year}} \times 1000$$

Is it a true rate?

$$q = \frac{e}{P_0}$$

A rate is not a probability

$$\bar{P} = \frac{P_0 + P_1}{2}$$

The mid-year population